

BillX GPS — Advanced Features

Client overview · Same logic on Mobile App & Web · Beyond basic live map

1-page brief
Effort = build + harden + test (hrs)

Basic GPS shows where a vehicle was last seen. **BillX advanced layer** shows how it is moving now, whether the signal is healthy, trip progress on assigned routes, replayable history, plus geofences, alerts, and commands — with smooth motion even when GPS updates are sparse.

1) LIVE TRACKING EXPERIENCE

Feature	Platform	Why advanced	Hrs
Follow Mode	App + Web	Camera stays locked on the moving vehicle — not a one-time “center map”.	40–70
Tracking HUD	Web / App	Live speed, status, heading, GPS quality, address & last update while following.	30–50
Smooth motion + dead reckoning	App + Web	Icon keeps moving between GPS packets (critical at highway speeds).	80–140
Realtime push (WebSocket)	App + Web	Locations pushed live; auto-fallback to polling if the socket drops.	80–130
Street View + Traffic	Web	Street-level view synced to the vehicle; live road congestion on the map.	90–135

2) SMART STATUS (IDENTICAL ON APP & WEB)

Feature	Platform	Why advanced	Hrs
Status engine	App + Web	Running / Idle / Moving / Parked from ignition + speed; Delayed / Stale / Offline from GPS age. Offline ≠ ignition off.	60–100
Status duration	App + Web	Shows how long the current state has lasted (needs transition history).	24–40

3) HISTORY, ROUTES & FLEET OPERATIONS

Feature	Platform	Why advanced	Hrs
History playback	App + Web	Play / pause / scrub trip history like a video — not only a static line.	100–160
Lite / fast history	App + API	Loads a light route first so multi-day history does not freeze phones.	30–50
Route trip progress + smart matching	App + Web	% complete, remaining km, ETA; on-route / off-route / arrived logic.	190–330
Geofences + smart alerts	App + Web	Zones with enter/exit; rule-based alerts via push / email / WhatsApp.	170–300
Remote commands + odometer	App + Web	Send tracker commands with status lifecycle; GPS odometer with client baseline.	120–200
Mobile performance stack	App	Overlay marker + throttled camera so follow mode stays smooth on mid-range Android.	180–320

Why clients call this “advanced”

Basic trackers answer “where?” BillX also answers “how is it moving?”, “is GPS healthy?”, “is the trip on plan?”, and “what happened on the route?” — with the same rules on phone and browser.

How to use the hour ranges

Hours are engineering effort (design, build, harden, test), not calendar days. Parallel work and shared backend reduce delivery time. Ranges reflect complexity, not a fixed quote.

Live + status core ~300–500 hrs	History + playback ~200–350 hrs	Routes + ops ~480–830 hrs	Mobile performance ~180–320 hrs
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